

For this project I would like to create a data pipeline that can help prepare data for economic forecasting.

This pipeline should check it's data before it is ingested for use.

Economic Forecasting
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Who will get value from this?

1 Work in risk management

2 Work in economic research and forecasting

3 Work in investing and asset management

4 Work in consumer spending

5 Like to trade on the side

6 Want to plan for the future

Data Inputs/Sources



Stock Data

This can be pulled through API's and is useful for gauging expectations in the economy



Commodity Data

This can come from API's websites and other tools. This can help see if there is a market shock and can help with understanding expectations.



Geopolitical Data

This can be received from government and private entities. This would be a hard data point to use since it is so complex, But it helps understand structural risk,



Government Spending

The government gives a lot of information through agencies. They are a huge spender and supporter for the economy.



Consumer Behavior

This helps fill the gap the other tools may have. This can be pulled from private institutions,

Three Phases for the Project

1

Data Ingestion

This will be the part of the tool that will gather the data from various source.s.
Each team member can focus on a type of data.

2

Data Validation

Data test will be created to confirm that this data is correct and is properly flowing through the pipeline.

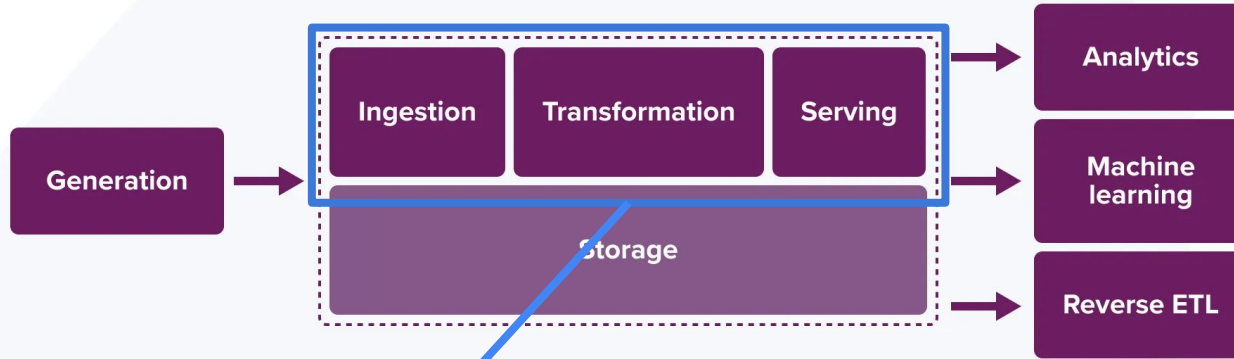
3

Data Serving

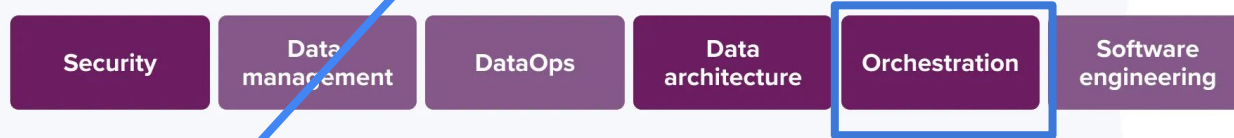
The data will need to be available for usage in the correct format.

Note: If we want we can break the project up into these three parts and have one person over each part.

Data engineering lifecycle



Undercurrents:



This project will focus on ingesting, transforming and serving data for usage.

Value of Data Engineering

- High-quality data is essential for producing accurate estimations.
- Flexibility is crucial, as economics is constantly evolving and changing.
- Risk reduction is a highly technical field that prioritizes precision and accuracy.
- Supports effective data organization and ensures information remains current and up to date

Challenges and Risk

- **Data ingestion:** Sourcing and reliably pulling data from multiple disparate APIs and reports, some of which may be unstructured or inconsistently formatted
- **Cross-source integration:** Reconciling data from sources with different formats, update cadences, and quality standards is one of the most technically demanding aspects of this project
- **Validation logic:** Designing checks that can catch both obvious errors and subtle inconsistencies that could compromise the integrity of the final dataset
- **Orchestration:** Ensuring that all pipeline stages are properly sequenced, automated, and monitored so the system runs reliably over time

Additional Thoughts

For this project I want the output to be a pipeline that can be ran that can give an output of values based off of economic inputs.

The outputs should be numbers of some sort: percentages, points, etc. Once these values are assigned an app should be able to pick this up and use it however it wants.

Note: I haven't picked which file format I want to use.

Two approaches I have for this project:

Team members can focus on individual inputs and making them digestible

Or

Team members can focus on a part of the data process mentioned in slide 4